

Mingtai Zhang

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Education

Guangxi University | School of Resources, Environment and Materials

2022 – 2026

B.Eng. in Materials Science and Engineering

Coursework: fundamentals of materials science, physical metallurgy, principles of metallurgical processes, materials characterization, physical chemistry, metal forming technology, mechanical design.

Research Summary

My thesis grows a lithiophilic magnesium interlayer on lithium metal by electrochemical pre-deposition; I assemble the cells, run the cycling, and do the XRD and EIS analysis myself. To verify how the interlayer works, every claim had to survive three independent cell configurations. The constraint is what my computational work is about: how many measurements are needed to determine an answer, and whether a reported number means what it claims. I was the data and software author on an identifiability result in ptychography, a technique used to image electrode interfaces, and first author on a large-scale audit of machine-learning systems. I am applying for a PhD in computational materials where both sides are the same job.

Research Interests

Lithium-metal anode interfaces and electrochemical deposition | MXene-modified separators and ion transport | computational imaging and machine learning for materials characterization | high-throughput electrochemical screening and self-driving laboratories | waste-to-catalyst resource recovery

Publications and Manuscripts

Co-author lists for papers under double-blind review are withheld until notification, following the anonymity policy of each venue.

Materials and Electrochemistry

- [1] Guang Zhang, Dong Yang, Kewang Yang, Farwa Mushtaq, Meng Zhang, **Mingtai Zhang**, Jieshang Lu, Yucheng Wen, and Meinan Liu. “Modified Ether Electrolyte by Siloxane Additive with Abundant Si–O Bonds for High-Voltage Lithium Metal Batteries.” *Journal of Physics: Conference Series* **3165**, 012009 (2025); CEAMS 2025, Dalian [EI-indexed]. DOI: 10.1088/1742-6596/3165/1/012009.
- [2] **Undergraduate thesis**. “Electrochemical Deposition of Magnesium for Constructing High-Performance Lithium Metal Anodes.” Guangxi University, 2026. Advisor: Prof. Meinan Liu.

Measurement, Imaging, and Machine Learning

- [3] Yuchen Miao, **Mingtai Zhang**, Zijun Wang, Ruobing Han, Chang Han, and Siyang Xu. “Two-Scan Recovery of Sparse Defects in Ptychography.” **Second author**; under review, *IEEE Signal Processing Letters*, 2026. (CRediT: data curation, software.)
- [4] Paper on attribution boundaries in conversational AI memory (title withheld during review) **First author**; under double-blind review, a top-tier HCI conference.
- [5] Paper on the accuracy–evaluation trade-off in test-time selection (title withheld during review) **Second author**; under double-blind review, a top-tier machine-learning conference.
- [6] Yuchen Miao, Zijun Wang, Chang Han, Yurui Shi, **Mingtai Zhang**, and Siyang Xu. “The ‘10th Juror’: Open-Set Standpoint Screening for Bureaucratic Bias Detection.” *EMNLP 2026* (Main Conference) [CORE A*, CCF-B].

Research Experience — Materials and Electrochemistry

Lithium-Metal Battery Group (Prof. Meinan Liu), Guangxi University | Undergraduate Researcher

2025 – present

Lithiophilic Mg interlayers by electrochemical pre-deposition | thesis project, sole author

- Designed a Li|Mg-electrolyte|Mg bimetallic CR2025 cell that plates a lithiophilic Mg layer straight onto Li foil. Current density and areal capacity were fixed so that the pre-cycling count (20/30/40) was the only variable. All assembly was done in an Ar glovebox below 0.1 ppm H₂O and O₂.
- Found the optimum at 30 cycles and identified a different failure mode on each side of it. At 20 cycles the Mg coverage is discontinuous and bare-Li nucleation sites remain, so efficiency collapses at cycle 45. At 40 cycles the alloy layer is too thick, cracks, and pushes the overpotential to 180 mV, with collapse at cycle 30.
- Li plating plateau overpotential dropped from 145 to 114 mV at 1 mA cm^{−2}, and the nucleation dip from about 30 to about 20 mV. Mg@Li-30 held about 100% Coulombic efficiency against Cu for 120 cycles, where bare Li fell from 98% to roughly 80%. Symmetric cells stayed flat near 30 mV for 330 h while bare Li drifted from 80 to about 180 mV.
- Checked the result at device level: Mg@Li-30||LFP full cells with a commercial 9.9 mg cm^{−2} cathode kept 85.3% capacity after 600 cycles at 0.5 C and gave about 145 mAh g^{−1} at 3 C, 1.45 times bare Li. EIS traced the gain to interfacial impedance falling from roughly 30 to 10 Ω, and XRD showed metallic Mg plus a bcc Li–Mg solid solution, which explains the lower nucleation barrier. The route uses only commercial electrolyte and standard CR2025 hardware, so other groups can repeat it.

Ti₃C₂T_x MXene-modified separators for dendrite suppression

- Run the MXene line end to end: selective etching of Al from the Ti₃AlC₂ MAX phase, intercalation and delamination into few-layer colloidal nanosheets, then film coating onto polyolefin separators.
- The idea is to make the separator do work rather than just separate. Anionic surface terminations pin anions and raise the Li⁺ transference number, so the plating flux reaching the anode is more uniform. SEM, XPS, and galvanostatic cycling link termination chemistry to transport and to cycle life.

Siloxane additives for high-voltage ether electrolytes | JPCS 2025 (EI)

- Contributed cell fabrication and electrochemical testing to the study that added decamethylcyclopentasiloxane (DMCPS) to suppress oxidative decomposition of ether electrolytes above 4 V. Retention rose from 70.8% to 73.9% over 700 cycles (0.5 C, 2.6–3.8 V), and Li||NCM811 from 70% at 20 cycles to 75% at 100 cycles (3.0–4.3 V).

- Turned the mixed heavy-metal cations in industrial wastewater into an entropy-stabilized single-phase spinel (AB_2O_4), so a waste stream becomes catalyst feedstock instead of sludge. I ran synthesis, phase control, and the photocatalytic VOC tests. Spinel high-entropy oxides and sludge-derived catalysts each exist separately; combining them for VOC oxidation is still open, and the non-equimolar cation mix that real wastewater gives you is a natural case for active-learning composition search.

Research Experience — Computation for Physical Measurement

The work above rests on inference about interfaces and data. The work below is about when that kind of inference is valid, and what it costs to check.

Compressive Ptychography: How Few Scans Suffice?

2026

Second author, IEEE SPL (under review) | forward-model simulation and the full evaluation harness

- Handled data curation and software for an inverse-problem study in coherent diffractive imaging, where the object is a known uniform background carrying a few unknown defects. That is the same problem as sparse defect metrology on wafers, thin films, and coatings.
- Built the forward-model simulation and the evaluation harness: Gaussian probes, scan geometries, sparse-defect ensembles, and detector noise. Benchmarked the proposed $2N$ -lag decoder against full-data linearization and the iterative phase-retrieval baselines PIE and HT-WF over defect-separation and SNR sweeps, with hundreds of trials per operating point.
- Working on an identifiability result changed how I read my own lab data. Before asking which method reconstructs better, I now ask whether the measurement can determine the answer at all.

Large-Scale Computational Experiments

2025 – present

First author, under double-blind review | second author, under double-blind review | evaluation-set construction, EMNLP 2026 Main

- Led a large-scale audit of machine-learning systems. I wrote down a falsifiable requirement, split it into two independent measures, and tested it over hundreds of controlled episodes, tens of thousands of scored outputs, and a large factorial of generated records, with a second model family as a replication arm and a held-out surface form to bound the scope. The result was negative and I reported it that way: two training interventions improved one output channel while making the other worse.
- Second author on a theory and experiment paper on test-time selection, which asks how much accuracy is reachable once you also have to prove that accuracy from a fixed pool of reused labels. Selecting better answers concentrates the output distribution and makes it harder to measure; the paper works out that trade-off and tests the resulting selection policies on four frozen reasoning banks.
- Contributed evaluation-set construction and analysis to an EMNLP 2026 main-conference paper on evidence-grounded agent pipelines ($F1 = 0.880$, 20.2 points above the strongest zero-shot baseline).
- Factorial design, thousands of scored trials, replication arms, and stated validity limits are the same machinery that high-throughput electrochemical screening and self-driving laboratories need. That is the part of my computational work I most want to bring back into materials research.

Industry and Engineering Experience

Four supervised placements, sixteen manufacturing enterprises across Guangxi

2024 – 2026

- Followed the aluminium chain end to end: open-pit bauxite mining and ore blending, Bayer digestion, red-mud settling and calcination, molten-salt electrolysis to primary aluminium, then casting, extrusion, and sheet and foil rolling. Sites included **Guangxi Huayin Aluminium** (Chalco JV, Debao), the **Chalco Guangxi Branch** at Pingguo, **Guangxi Huasheng New Material**, and **Guangxi Guangtong Liuzhou Aluminium**.
- **Closest to my own research:** at the former **Guangxi Nannan Aluminium Foil** works, now **Nanning Industrial Investment Aluminium-Based New Materials Group**, I followed the full ingot-to-foil chain for lithium-ion battery aluminium foil, down to $12\ \mu\text{m}$ double-side-bright 1xxx-series cathode current-collector foil, carbon-coated foil, and battery-case stock shipped to EV battery lines.
- Also covered integrated steelmaking and stainless coil at **Guangxi Liuzhou Iron and Steel** and **Guangxi BG New Materials**; vehicle stamping, welding, painting, and assembly at **Dongfeng Liuzhou Motor**; heavy structural fabrication at **LiuGong Machinery**; crankshaft die forging at **Guilin Fuda**; high-pressure high-temperature sintering of PDC and PCBN at **Guilin Tebang**; sand casting at **Guangxi Yuchai Foundry**; and photovoltaic module lines at **Guangxi Dasheng New Energy** and **Guangxi Jingfei Energy**.

Entrepreneurship, Leadership, Honors, and Technical Skills

Ventures: Founder, Kangjie returning-student tutoring platform, Yuncheng, Shanxi (winter 2026) — non-profit service staffed by university students home for the holiday; I ran scheduling, tutor-student matching, and progress tracking, and automated most of the coordination with AI agent tooling; in total the service mobilized 28 tutors to serve 37 high-school students. Co-founder, on-demand fitness coaching platform with the Guangxi University Rehabilitation Association (2026 – present), now in live operation; I own operating strategy and resource coordination.

Leadership: Captain, Guangxi University Football Team | President, University Football Association | President, Student Union, School of Resources, Environment and Materials.

Honors: Silver Award, Guangxi University Career Planning Competition (2023) | Third Prize, Guangxi University Metallography Contest (2023) | First Place (2025) and Third Place (2024), Guangxi “Qianli Cup” Intercollegiate Football League, undergraduate division.

Electrochemistry: coin-cell assembly, Ar glovebox, CV, EIS, electrochemical workstations | **Synthesis:** electrochemical deposition, MXene etching, intercalation, delamination and coating, high-entropy oxides, electrode and cell fabrication | **Characterization:** XRD, SEM, XPS |

Computation: Python, PyTorch, $\text{\LaTeX}$, Origin; factorial design, large-scale evaluation harnesses, forward-model simulation, statistical analysis | **Process engineering:** Bayer refining, electrolysis, casting, extrusion, rolling, forging, foundry, SOP and laboratory safety